Linear Algebra - CS - 3A & 3B

Problem Sheet: Orthogonality and Least Square Method

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A Orthogonality and Gram-Schmidt Process

1. **(Basic Orthogonality Check in $\mathbb{R}^2$)** Determine whether the following pairs of vectors are orthogonal:

$$\text{(a) } \mathbf{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \qquad \text{(b) } \mathbf{u} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

2. **(Orthogonal Set in $\mathbb{R}^3$)** Show that the set

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix} \right\}$$

is orthogonal. Then convert it to an orthonormal set.

3. **(Gram-Schmidt in $\mathbb{R}^2$)** Apply the Gram-Schmidt process to the basis:

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

to produce an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2\}$.

4. **(Gram-Schmidt in $\mathbb{R}^3$ - Independent Vectors)** Apply the Gram-Schmidt process to:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Find the orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$.

5. **(Gram-Schmidt with Dependent Vectors)** Apply the Gram-Schmidt process to:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

   What happens at step 2? Explain.

6. **(Conceptual)** Explain why the Gram-Schmidt process fails when the input vectors are linearly dependent. What does this tell us about the relationship between orthogonality and linear independence?

B Least Square Problems

1. **(Basic Least Squares - Line Fitting in $\mathbb{R}^2$)** Find the least squares line $y = mx + b$ that best fits the data points:

$$(1,2), \quad (2,3), \quad (3,5), \quad (4,6)$$

   Predict the value of y when $x = 5$.

2. **(Least Squares with QR Decomposition)** Given the system $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

   (a) Find the QR decomposition of A using Gram-Schmidt.

   (b) Use the QR decomposition to solve the least squares problem.

   (c) Verify that the residual $\mathbf{r} = \mathbf{b} - A\mathbf{x}$ is orthogonal to the columns of A.

3. **(Quadratic Curve Fitting)** Fit a quadratic polynomial $y = ax^2 + bx + c$ to the following data:

$$(-2,1), \quad (-1,0), \quad (0,1), \quad (1,4), \quad (2,9)$$

   Find the coefficients a, b, c and compute the sum of squared errors.

4. **(Overdetermined System with 3 Equations, 2 Unknowns)** Solve the following overdetermined system using least squares:

$$\begin{cases} x + 2y = 5 \\ 2x + y = 6 \\ 3x + 2y = 8 \end{cases}$$

   Set up the normal equations $A^T A\mathbf{x} = A^T\mathbf{b}$ and solve for x and y.

5. **(Real-World: Temperature Prediction)** A scientist records temperature at different times:

Time (hours)	Temperature (°C)
0	10
1	12
2	15
3	14
4	18

   Find the best-fit line $T = mt + b$ and predict the temperature at $t = 5$ hours.

6. **(Conceptual)** Explain the geometric interpretation of the least squares solution. Why is the residual vector orthogonal to the column space of A? Include a diagram description in your answer.